

Use Case 1: AI-assisted skin lesion diagnosis



Dermatofibroma: 0.6
Melanocytic nevus: 0.2
Melanoma: 0.1
Basal cell carcinoma: 0.0
Actinic keratosis: 0.0
Benign keratosis: 0.0
Vascular lesion: 0.1

Problem category: Image Level Classification

Recommended metrics:

- Multi-class Counting Metric: [Balanced Accuracy \(BA\)](#)
- Per-class Counting Metric: [Positive Likelihood Ratio \(LR+\)](#)
- Multi-threshold Metric: [Area under the Receiver Operating Characteristic Curve \(AUROC\)](#)
- Calibration Metrics: [Expected Calibration Error \(ECE\)](#) and [Root Brier Score \(RBS\)](#)

Fingerprint answers:

Fingerprint	Answer
Decision rule (cutoff method) applied to predicted class scores	Argmax-based
Penalization of errors: Unequal severity of class confusions	No
Penalization of errors: Unequal interest across classes	No
Provided class prevalences reflect the population of interest	No
Availability of predicted class scores	Yes
Calibration assessment requested	Yes
Comparative calibration assessment requested	No comparative assessment
Interpretable estimate of calibration error requested	Assessment of calibration error in isolation

Decision between metrics:

- [Multi-class Counting Metric] Balanced Accuracy (BA) vs. Expected Cost (EC)
 - Answer: Balanced Accuracy (BA)
- [Per-class Counting Metric] Positive Likelihood Ratio (LR+) vs. Sensitivity
 - Answer: Positive Likelihood Ratio (LR+)
- [Calibration Metric]: Expected Calibration Error (ECE) & Root Brier Score (RBS) vs. Classwise Calibration Error (CWCE) vs. Expected Calibration Error Kernel Density Estimate (ECE^{KDE}) & Class-wise Calibration Error (CWCE) & Root Brier Score (RBS)
 - Answer: ECE (top-label error) and RBS